

The Coarse-Geometric Karchmer-Wigderson Programme: A scaffold for formula-depth lower bounds via asymptotic dimension and the Roe pairing

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Abstract

We propose a research programme — not a result — to attack formula-depth lower bounds via the language of large-scale geometry and the Roe-style coarse cohomology of the Karchmer-Wigderson pair space. Concretely: for a boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$, let $X_f = f^{-1}(0) \times f^{-1}(1)$ equipped with the disagreement- Hamming pseudo-metric d_f . We define a coarse depth invariant $\kappa(f)$ via the Roe-trace pairing of 1-cocycles against controlled operators on X_f , and outline five tentative axioms whose combination would give a κ -based formula depth lower bound that is, by construction, immune to the natural-proofs, relativization, and algebrization barriers. We make explicit why this is currently a *programme*: ten sub-conjectures generated under the framework have all returned **INCONCLUSIVE** (test crashes during verification) or **BARRIER_HIT** verdicts, none have been empirically supported, and no lean-verified theorem is yet attached. We provide a Lean 4 stub that records the primitives and axioms in machine-readable form and lay out the explicit open problems that turn the programme into a discipline.

1 Why a programme, not a paper of theorems

Formula-depth lower bounds for explicit boolean functions are blocked by three barriers: *relativization* [?], *natural proofs* [?], and *algebrization* [?]. Almost every existing bound is sub-quadratic in formula size; the long-standing open problem is whether explicit functions admit super-logarithmic formula depth, which would separate **NC**¹ from **P**. The Karchmer-Wigderson characterization [?] expresses formula depth as the communication cost of a two-party game on the pair space X_f . Coarse geometry, in the sense of Roe [?] and Yu [?], provides invariants of metric spaces that:

1. are *coarsely invariant* (so they cannot distinguish a random f from a structured f at the Razborov-Rudich density level);
2. are *destroyed by oracle access* (oracle gates collapse d_f);
3. live in the K-theory of the Roe algebra $C_u^*[X_f]$, which is not closed under polynomial extensions of the underlying alphabet.

The first three properties are precisely those that the three barriers exclude. The CG-KW programme aims to construct a formula-depth lower bound from a coarse invariant that satisfies all three.

What we have: a precise definition of X_f , d_f , the controlled-cover and Roe-pairing primitives, a target invariant $\kappa(f)$, and five tentative axioms (§??).

*The CG-KW framework was first proposed by an autonomous LLM-driven proposer agent of the [Sperimen-talMath](#) research engine and refined under the five-gate review pipeline of `MULTIAGENT_PIPELINE.md`. The five primitives, five tentative axioms, and Lean 4 stub presented here are the framework’s current state; none of the axioms are proved, and the explicit-function targets remain open.

What we lack: a proof of any of the five axioms. The ten sub-conjectures generated under the framework are listed in §?? together with their current pipeline verdicts. None is currently supported.

2 Primitives

Throughout, $f: \{0, 1\}^n \rightarrow \{0, 1\}$ is a boolean function and $X_f = f^{-1}(0) \times f^{-1}(1)$.

Definition 1 (KW-metric space). *The KW-metric on X_f is*

$$d_f((x, y), (x', y')) = \log_2 \min\{|S| : S \subseteq [n], x|_S \neq y|_S \text{ certifies } f(x) \neq f(y) \text{ or } x'|_S \neq y'|_S \text{ certifies } f(x') \neq f(y')\}$$

Definition 2 (Controlled cover). *A finite family $\mathcal{U} = \{U_i\}_{i \in I}$ of subsets of X_f with diameter bound R and multiplicity bound m : every U_i has d_f -diameter $\leq R$ and every $p \in X_f$ belongs to at most m of the U_i .*

Definition 3 (Coarse 1-cocycle). *A function $c: X_f \times X_f \rightarrow \mathbb{Z}$ of finite support modulo controlled equivalence, satisfying the cocycle identity $c(p, q) + c(q, r) = c(p, r)$ on every triple. Class in $HX^1(X_f)$ in the Roe-style coarse cohomology.*

Definition 4 (Roe-controlled operator). *A bounded operator $T \in C_u^*[X_f]$ with finite propagation $\text{prop}(T) := \sup\{d_f(p, q) : T_{pq} \neq 0\}$.*

Definition 5 (Asymptotic-dimension witness). *A vertex coloring $\chi: X_f \rightarrow [k]$ together with a diameter bound R such that each colour class decomposes into d_f -bounded pieces of diameter $\leq R$ that are pairwise R -disjoint. Witnesses $\text{asdim}(X_f) \leq k - 1$ in the Gromov sense.*

3 The target invariant $\kappa(f)$

Definition 6 (Coarse depth).

$$\kappa(f) := \sup_{\substack{c \in HX^1(X_f), T \in C_u^*[X_f] \\ \langle c, T \rangle \neq 0}} \frac{\log |\langle c, T \rangle|}{\log \text{prop}(T)},$$

where $\langle c, T \rangle$ is the Roe-trace pairing $\sum_{p, q} c(p, q) \cdot T_{pq}$ with appropriate normalization.

The programmatic claim is that κ tracks formula depth.

4 The five tentative axioms

Conjecture 1 (A1, Coarse-KW link). *There is a constant $c_0 > 0$ such that for every boolean f ,*

$$d(f) \geq c_0 \cdot \kappa(f),$$

where $d(f)$ is the De Morgan formula depth.

A1 is the load-bearing axiom; everything else is a sufficient condition for κ to be a non-trivial invariant on explicit functions.

Conjecture 2 (A2, Composition sub-additivity). *For all f, g and the block composition $f \circ g^n$,*

$$\text{asdim}(X_{f \circ g^n}) \geq \text{asdim}(X_f) + \text{asdim}(X_g) - O(1).$$

A2 is a coarse-geometric analogue of the Karchmer-Raz-Wigderson direct sum theorem [?].

Conjecture 3 (A3, Anti-natural-proofs). *For a uniformly random f , $HX^1(X_f) = 0$ generically. Consequently the predicate “ $\kappa(f) \geq T$ ” is not a largeness property in the Razborov-Rudich sense.*

Conjecture 4 (A4, Anti-relativization). *κ is a metric invariant of d_f . Oracle access collapses d_f to ≤ 1 , hence κ to ≤ 1 , hence cannot relativize.*

Conjecture 5 (A5, Roe-index rigidity). *Nontrivial classes in $HX^1(X_f)$ detected by the Roe-trace pairing are stable under coarse equivalence (in the spirit of Yu’s coarse Baum-Connes result [?] and Nowak-Yu’s compendium [?]). Andreev’s function [?] and element-distinctness are conjectured to realize $\kappa = \Omega(\log^2 n)$.*

5 Programme-level explicit-function targets

Problem 1 (Andreev’s function). *Prove $\kappa(\text{Andreev}_n) = \Omega(\log^2 n)$. Combined with A1, this yields a super-logarithmic formula depth lower bound on Andreev_n , separating NC^1 from $\mathbf{P}$ on this family.*

Problem 2 (Indexing function). *Prove $\kappa(\text{IND}_k) \geq k - O(1)$ where $N = k + 2^k$. The formula depth of IND_k is $\Theta(k)$, so this would match the true depth up to a constant via A1.*

Problem 3 (Element-distinctness). *Prove $\kappa(\text{ED}_n) = \Omega(\log^2 n)$.*

6 Status of generated sub-conjectures

The autonomous engine has generated 10 sub-conjectures under the CG-KW framework (entry IDs in the [notebook](#)). None are currently supported.

Entry ID	Verdict	Subject
e871be18b26c	INCONCLUSIVE	asdim lower bound for IND_k
738bb754804a	INCONCLUSIVE	Roe-trace pairing for Inner-Product
7c5ad65991c5	INCONCLUSIVE	$HX^1(X_f) = 0$ for random f (A3 test)
f685d342f4c0	INCONCLUSIVE	KRW direct sum at the asdim level (A2 test)
a81b4af0bc2b	INCONCLUSIVE	Oracle collapse of κ (A4 test)
cc8127953d91	INCONCLUSIVE	Pullback monotonicity under coarse-Lipschitz reductions
ed6023df7533	BARRIER_HIT	Diameter-multiplicity power law (NATURAL_PROOFS)
2ee2e2f64a25	BARRIER_HIT	Monotone ultralimit of κ (NATURAL_PROOFS)
8047f7f8eea1	INCONCLUSIVE	Block-composition sub-additivity via Roe-trace pairing
264c40c0d040	INCONCLUSIVE	Coboundary-vanishing threshold HX^1

Table 1: Current pipeline verdicts. Most INCONCLUSIVE entries have runtime errors in their auto-generated empirical tests (e.g. $\log_2(0)$ domain errors); they are queued for repair under the new five-gate pipeline.

Remark (An internal tension to resolve). *Two sub-conjectures (ed6023df7533, 2ee2e2f64a25) were flagged BARRIER_HIT on the natural-proofs barrier despite axiom A3 of the framework being explicitly an anti-natural-proofs claim. This is currently a contradiction: either (a) the proposer’s barrier filter incorrectly flags A3-compatible statements; (b) those particular sub-conjectures violate A3 in a way the framework’s verbal description did not capture; or (c) A3 is too strong as stated. Resolving this tension is the first concrete task on the CG-KW programme’s to-do list.*

7 Lean 4 record

The accompanying file [lean_verified/cg_kw/CGKW.lean](#) records the primitives, the target invariant, and the five axioms in machine-readable form. `lake build` succeeds; the file does not depend

on Mathlib, uses `Float` as a stand-in for $\mathbb{R}$, and contains *no proofs*. Each axiom of §?? is recorded as a Lean `axiom`, acknowledging that it is open. Each explicit-function target of §?? is recorded as a `def` of type `Prop`.

A Mathlib port (lifting every `Float` to `Real` and every `axiom` to a `theorem` with `sorry`) is queued.

8 What turns this scaffold into a paper of theorems

The CG-KW programme converts the long-standing problem of formula-depth lower bounds into seven concrete tasks of independent interest:

1. Prove the Coarse-KW link (A1) for at least one nontrivial class of boolean functions.
2. Prove the asdim sub-additivity (A2), even with a $\log n$ slack.
3. Prove the anti-natural-proofs property (A3) by establishing $HX^1(X_f) = 0$ generically — a Roe-algebraic concentration of measure result.
4. Prove the anti-relativization property (A4) — likely the shortest of the five proofs.
5. Construct an explicit cocycle-operator pair (c, T) on X_{Andreev} realizing $\kappa = \Omega(\log^2 n)$.
6. Repair the ten generated sub-conjecture tests (§??) and re-run them through the five-gate pipeline.
7. Resolve the A3 tension noted above.

A successful resolution of any single one of (1)–(5) would constitute a publishable paper in its own right. The scaffold itself is presented here in the format of Mulmuley-Sohoni’s GCT programme [?]: a proposed direction with explicit, testable, sub-conjectures, not a claim of a result.

Reproducibility. The Lean stub builds in ~ 500 ms with `lake build`; the sub-conjecture data is auto-synced hourly to github.com/ludwigkubler/SperimentalMath.

References

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